

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Uses an appropriate method for ensuring the line lies in the plane	AO3.1a	M1	Let $\lambda = x - p = \frac{y+2}{q} = 3 - z$
	Obtains equation(s) in p and q	AO1.1a	M1	then $x = \lambda + p, y = q\lambda - 2, z = 3 - \lambda$ sub into equation of plane $(\lambda + p) - (q\lambda - 2) - 2(3 - \lambda) = -3$ $\lambda(3 - q) + (p - 1) = 0$ this is true for all λ therefore $p = 1$ and $q = 3$
	Deduces the values of p and q	AO2.2a	A1	ALT vector equation of line is $\mathbf{r} = \begin{pmatrix} p \\ -2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ q \\ -1 \end{pmatrix}$ $\begin{pmatrix} p \\ -2 \\ 3 \end{pmatrix}$ lies on the plane $\begin{pmatrix} p \\ -2 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -1 \\ -2 \end{pmatrix} + 3 = 0$ $\begin{pmatrix} 1 \\ q \\ -1 \end{pmatrix}$ And $\begin{pmatrix} 1 \\ q \\ -1 \end{pmatrix}$ is $\begin{pmatrix} 1 \\ -1 \\ -2 \end{pmatrix}$ perpendicular to

			$\begin{pmatrix} 1 \\ q \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -1 \\ -2 \end{pmatrix} = 0$ therefore $\Rightarrow q = 3 \text{ and } p = 1$
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(b)	States that to have a solution the coefficient of λ in equation from (a) cannot be 0 OR dot product must $\neq 0$	AO2.4	R1	$\begin{pmatrix} 1 \\ q \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -1 \\ -2 \end{pmatrix} \neq 0 \Rightarrow q \neq 3$
	Deduces correct range of values for q	AO2.2a	R1	
	Deduces the range of values for p	AO2.2a	R1	p can take any value
(c) (i)	Finds the correct scalar product of the normal to the plane and the direction vector	AO1.1b	B1	$\mathbf{n} = \begin{pmatrix} 1 \\ -1 \\ -2 \end{pmatrix} \quad \mathbf{d} = \begin{pmatrix} 1 \\ q \\ -1 \end{pmatrix}$
	Correctly deduces the value of $\cos \alpha$	AO2.2a	R1	$\mathbf{n} \cdot \mathbf{d} = 3 - q$
	Forms an equation connecting all relevant parts using $\mathbf{n} \cdot \mathbf{d} = \mathbf{n} \mathbf{d} \cos \theta$	AO3.1a	M1	Let α be angle between the line and the normal to the plane $\sin \theta = \frac{1}{\sqrt{6}} \Rightarrow \cos \alpha = \frac{1}{\sqrt{6}}$
	Obtains correct value for q	AO1.1b	A1	$q - 3 = \sqrt{6} \sqrt{q^2 + 2} \times \left(\frac{1}{\sqrt{6}} \right)$ $(3 - q)^2 = q^2 + 2$ $\Rightarrow 6q = 7 \text{ giving } q = \frac{7}{6}$
(ii)	Uses 'their' expressions for x and y and 'their' value for q and the equation of the plane to form an equation to find p	AO3.1a	M1	$x - p = \frac{y + 2}{\frac{7}{6}} = 3 - z$ $z = 0 \Rightarrow x = p + 3, = 1.5$
	Uses $z = 0$ to deduce expressions for x and y in terms of p and q	AO2.2a	R1	$\begin{pmatrix} p + 3 \\ 1.5 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -1 \\ -2 \end{pmatrix} = -3$
	Obtains the correct value of p CAO	AO1.1b	A1	$\Rightarrow p + 3 - 1.5 = -3$ $\Rightarrow p = -4.5$

Q2.

Marking Instructions	AO	Marks	Typical Solution
Uses the mathematical model to find the volume by first finding the coordinates of A. To award this mark must see an attempt to find coords of A and an attempt at volume of prism.	AO3.4	M1	$x = 4t - 4$ $y = 12 - 12t$ $z = 4$ $4t - 4 - 3(12 - 12t) = 0$
Selects method involving both equation of plane and equation of line to find coords of A. Either using parametric form or using cross product Ignore sign errors	AO3.1a	M1	$40t - 40 = 0$ $t = 1$ $(0 \ 0 \ 4)$ OR
Either collects terms together and solves to find value of parameter for 'their' equation Or correctly calculates cross product for 'their' vectors	AO1.1b	A1F	$\begin{bmatrix} x \\ y \\ z \end{bmatrix} - \begin{bmatrix} -4 \\ 12 \\ 4 \end{bmatrix} \times \begin{bmatrix} 4 \\ -12 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$
Deduces the correct coordinates of A	AO2.2a	A1	$\begin{pmatrix} 3y+4 \\ y-12 \\ z-4 \end{pmatrix} \times \begin{pmatrix} 4 \\ -12 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$ $12(z - 4) = 0 \Rightarrow z = 4$
Selects a correct approach to calculate the volume of the prism.	AO3.1a	M1	$-12(3y + 4) - 4(y - 12) = 0$ $\Rightarrow y = 0, x = 0$ A has coordinates (0,0,4)
Finds two sides of the triangle ABC in vector form FT 'their' A	AO1.2	A1F	$\vec{AB} = \begin{pmatrix} 3 \\ 1 \\ 7 \end{pmatrix}$

Finds area of ABC FT 'their' A	AO1.1b	A1F	$\vec{AC} = \begin{pmatrix} 9 \\ 3 \\ 0 \end{pmatrix}$
Finds length of prism FT 'their' A	AO1.1b	A1F	$\vec{AB} \times \vec{AC} = \begin{pmatrix} -21 \\ 63 \\ 0 \end{pmatrix}$
Gives their answer in context by correctly finding the volume of the roof with correct units. FT 'their' prism	AO1.1b	A1F	$\text{Area } ABC = \frac{21\sqrt{10}}{2}$ $d = 4\sqrt{10}$ $\text{Volume} =$ $V = \frac{21\sqrt{10}}{2} \times 4\sqrt{10} = 420\text{m}^3$
Total 9 marks			

Q3.

(a) (i) $\vec{AB} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$ $\vec{AD} = \begin{bmatrix} 4 \\ 3 \\ -1 \end{bmatrix}$
 either $\vec{AB}$ or $\vec{AD}$

B1

$$\begin{vmatrix} \mathbf{i} & 2 & 4 \\ \mathbf{j} & -1 & 3 \\ \mathbf{k} & 3 & -1 \end{vmatrix} = \begin{pmatrix} -8 \\ 14 \\ 10 \end{pmatrix}$$

one component of $\vec{AB} \times \vec{AD}$ correct

M1

all correct

A1cao

3

(ii) $\text{Area } ABCD = |\vec{AB} \times \vec{AD}| = \sqrt{8^2 + 14^2 + 10^2}$
 FT their $|\vec{AB} \times \vec{AD}|$

M1

$$= \sqrt{64 + 196 + 100}$$

$$= \sqrt{360}$$

$$= \sqrt{36} \sqrt{10}$$

$$= 6\sqrt{10}$$

$$\text{or } p = 6$$

A1cso

2

- (b) $\overrightarrow{AB} \times \overrightarrow{AD}$ is perpendicular to plane $ABCD$
 used for $\mathbf{v}$ in vector line equation

M1

Hence direction ratios of line $= -8 : 14 : 10 = -4 : 7 : 5$

$$\text{or any multiple of } \begin{bmatrix} -4 \\ 7 \\ 5 \end{bmatrix}$$

A1

M is mid-point of either diagonal $= \left(\frac{1+7}{2}, \frac{0+2}{2}, \frac{2+4}{2} \right)$
 mid-point calculation

B1

$$= (4, 1, 3)$$

$$\text{Hence line is } \left(\mathbf{r} - \begin{bmatrix} 4 \\ 1 \\ 3 \end{bmatrix} \right) \times \begin{bmatrix} -4 \\ 7 \\ 5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{All correct or equivalent multiple of } \begin{bmatrix} -4 \\ 7 \\ 5 \end{bmatrix}$$

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4

- (c) Perpendicular vector to $\Pi = \begin{bmatrix} -4 \\ 7 \\ 5 \end{bmatrix}$

$\Rightarrow \Pi$ has equation $-4x + 7y + 5z = c$
 ft their perpendicular vector

M1

Through $(6, 5, 17) \Rightarrow c = -4(6) + 7(5) + 5(17)$
 using $(6, 5, 17)$

m1

$$= -24 + 35 + 85 = 96$$

A1

Equation is $-4x + 7y + 5z = 96$

ACF

$$x = 4 - 4t; y = 1 + 7t; z = 3 + 5t$$

parametric form of line

B1F

Line meets plane when

$$-4(4 - 4t) + 7(1 + 7t) + 5(3 + 5t) = 96$$

$$-16 + 16t + 7 + 49t + 15 + 25t = 96$$

$$90t = 90$$

$$t = 1$$

substitution of parametric form and attempt to solve for t

M1

$$\Rightarrow \text{point of intersection} = (0, 8, 8)$$

correct point of intersection

A1cao

6

Alternative 1:

$$\mathbf{r} = \begin{bmatrix} 6 \\ 5 \\ 17 \end{bmatrix} + s \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} + t \begin{bmatrix} 4 \\ 3 \\ -1 \end{bmatrix}$$

$\mathbf{r} = \mathbf{a} + s\mathbf{d}_1 + t\mathbf{d}_2$ fully correct

(M1)

$$\begin{vmatrix} i & 2+2s+4t & -4 \\ j & 4-s+3t & 7 \\ k & 14+3s-t & 5 \end{vmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Substitute in $(\mathbf{r} - \mathbf{a}) \times \mathbf{b} = 0$

(m1)

Use of $\mathbf{r} - \mathbf{a}$ in parametric form – simplified

(B1F)

$$-26s + 22t = 78$$

$$22s + 16t = -66$$

$$10s + 40t = -30$$

Three correct equations obtained from vector product-terms collected

(A1)

$$s = -3, t = 0$$

Correctly solving equations to get both parameters

(M1)

$$\begin{bmatrix} 0 \\ 8 \\ 8 \end{bmatrix}$$

Correct point of intersection

(A1)

(6)

Alternative 2:

$$\mathbf{r} = \begin{bmatrix} 6 \\ 5 \\ 17 \end{bmatrix} + s \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} + t \begin{bmatrix} 4 \\ 3 \\ -1 \end{bmatrix}$$

$\mathbf{r} = \mathbf{a} + s\mathbf{d}_1 + t\mathbf{d}_2$ fully correct

(M1)

$$\mathbf{r} = \begin{pmatrix} 4-4p \\ 1+7p \\ 3+5p \end{pmatrix}$$

Parametric form of line

(B1F)

$$2s + 4t + 4p = -2$$

$$-s + 3t - 7p = -4$$

$$3s - t - 5p = -14$$

Equating components, simplifying and attempting to solve-must at least reduce to 2 equations in two unknowns

(m1)

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 $p = 2$

Solving equations-one parameter correct

(M1)

$$t = 0 \text{ and } s = -3$$

All values correct

(A1)

$$\begin{bmatrix} 0 \\ 8 \\ 8 \end{bmatrix}$$

Correct point of intersection

(A1)

(6)

[15]

Q4.

(a)

$$\left. \begin{array}{l} \text{direction ratios of line} = p : 3 : -1 \\ \text{normal to plane} = 1 : 1 : 2 \end{array} \right\} \begin{array}{l} \text{not} \\ \text{equal} \end{array}$$

Accept not parallel or showing vector product is non zero

B1

1

(b) $x = 3 + pt$
 $y = q + 3t$
 $z = 1 - t$

Parametric form seen

M1

Meets plane $\Rightarrow 3 + pt + q + 3t + 2(1 - t) = 10$

$\Rightarrow (5 + q) + t(p + 1) = 10$

Correct substitution in plane

A1

Within plane $\Rightarrow q = 5, p = -1$

M1 – Finding one correct value

A1 – Both values correct

M1A1

4

Alternative:

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 Point $(3, q, 1)$ is common to line and plane

Hence $\begin{pmatrix} 3 \\ q \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = 10$ which gives $q = 5$

Uses common point to find q

(M1A1)

Another point common to both is $(3 + p, 8, 0)$

Hence $\begin{pmatrix} 3+p \\ 8 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = 10$ which gives $p = -1$

Use of second point and value of q to find p or

consideration of scalar product $\begin{pmatrix} p \\ 3 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$

(M1A1)

(4)

(c) (i) $\mathbf{n} = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} \quad \mathbf{d} = \begin{pmatrix} p \\ 3 \\ -1 \end{pmatrix}$

Let α be angle between normal and direction ratios
(plane) (line)

$$\mathbf{n} \cdot \mathbf{d} = p + 1$$

$\mathbf{n} \cdot \mathbf{d}$ correct

M1

$$\sin \theta = \frac{1}{\sqrt{6}} \Rightarrow \cos \alpha = \frac{\pm 1}{\sqrt{6}}$$

Correct $\cos \alpha$ stated or implied

B1

$$\Rightarrow \frac{p+1}{\sqrt{6}\sqrt{p^2+10}} = \frac{\pm 1}{\sqrt{6}}$$

Forming equation connecting all relevant parts **and**
attempting to solve for p (condone missing $\pm$)
Dependent on first M1 – fully correct for A1

m1A1

$$\Rightarrow (p+1)^2 = p^2 + 10$$

$$\Rightarrow p^2 + 2p + 1 = p^2 + 10$$

$$\Rightarrow 2p = 9 \text{ giving } p = 4.5$$

CAO

A1

5

Alternative:

$$|\mathbf{n} \times \mathbf{d}| = \sqrt{49 + (1+2p)^2 + (3-p)^2}$$

$|\mathbf{n} \times \mathbf{d}|$ correct

(M1)

$$\sin \theta = \frac{1}{\sqrt{6}} \Rightarrow \sin \alpha = \frac{\sqrt{5}}{\sqrt{6}}$$

Correct sin α stated or implied

(B1)

$$\frac{\sqrt{49 + (1+2p)^2 + (3-p)^2}}{\sqrt{6}\sqrt{p^2+10}} = \frac{\sqrt{5}}{\sqrt{6}}$$

Forming equation connecting all relevant parts **and** attempting to solve for p . Dependent on first M1 – fully correct for A1

(m1A1)

Leading to $p = 4.5$
CAO

(A1)

(5)

(ii) $z = 2 \Rightarrow t = -1 \Rightarrow x = -1.5$

$p = 4.5 \quad y = q - 3$

$\Rightarrow -1.5 + q - 3 + 4 = 10$

Attempt to form an equation for q using $t = -1$

M1

$q = 10.5$
CAO

A1

²

EXAM PAPERS PRACTICE [12]

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Q5.

(a) Use of scalar product on

$\begin{pmatrix} 2 \\ 1 \\ 7 \end{pmatrix}$ and $\begin{pmatrix} 3 \\ 1 \\ -4 \end{pmatrix}$

M1

Sc.Prod. = ± 21 ;

B1

Moduli $\sqrt{54}$ and $\sqrt{26}$ correct
Accept 7.348... & 5.099...

B1

AWRT 56°

A1

- (b) $2x + y + 7t = 10$ and $3x + y - 4t = 7$ noted or used

M1

Eliminating (say) y to get x as a fn. of t

M1

$$x = 11t - 3$$

CAO

A1

Subst^g. back for y

M1

$$y = 16 - 29t$$

CAO

A1

$$\frac{x+3}{11} = \frac{y-16}{-29} = z \quad (=t)$$

B1 ft

6

- (c) Attempt at either $\begin{pmatrix} \lambda+20 \\ 9\lambda-1 \\ 4\lambda+7 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 1 \\ 7 \end{pmatrix} = 10$ or $\begin{pmatrix} \lambda+20 \\ 9\lambda-1 \\ 4\lambda+7 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 1 \\ -4 \end{pmatrix} = 7$

M1

Solving either

$$40 + 2\lambda - 1 + 9\lambda + 49 + 28\lambda = 10 \text{ or}$$

$$60 + 3\lambda - 1 + 9\lambda - 28 - 16\lambda = 7$$

M1

$$\lambda_1 = -2$$

$$\lambda_2 = 6$$

A1A1

$$P = (18, -19, -1) \text{ and } Q = (26, 53, 31)$$

$$PQ = \sqrt{8^2 + 72^2 + 32^2} = \sqrt{6272} = 56\sqrt{2}$$

NB P, Q not required:

$$d = |\lambda_1 - \lambda_2| \times |\mathbf{i} + 9\mathbf{j} + 4\mathbf{k}| = 8 \times 7\sqrt{2} = 56\sqrt{2} \quad \mathbf{M1 \ A1}$$

Q6.

- (a) For attempt at / getting normal d.v. $\begin{bmatrix} 1 \\ 5 \\ 7 \end{bmatrix}$

M1
A1

For getting a point on the line
eg $(0, -8, -11)$

M1
A1

For line equation: $(\mathbf{r} =)$ (their point) + λ (their d.v.)
Condone lack of r or r here

B1

ALT
eg Adding Π_1 and Π_2

(M1)

$$\Rightarrow 5x - y = 8$$

(B1)

Setting (eg) $x = \lambda$ and getting y, z in terms of λ :
 $y = 5\lambda - 8, z = 7\lambda - 11$

(M1)

Turning this into a vector equation of the given form

(M1)

$$(\mathbf{r} =) \begin{bmatrix} 0 \\ -8 \\ -11 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 5 \\ 7 \end{bmatrix}$$

(A1)

5

- (b) (i) Substituting their x, y, z in terms of λ into Π_3 's equation
($12x - y - z = 40$)

M1

For correct statement with no λ 's in: ($19 = 40$)



ft on incorrect values from their "point"

A1

Correct conclusion, from valid working, that the system is inconsistent

or "consistent" if it genuinely yields $40 = 40$

B1

3

- (ii) The three planes form a (triangular) prism – allow clear diagram

ft "sheaf" from a "consistent" conclusion

B1

1

- (c) (i) e.g. $\Pi_2 + \Pi_3 \Rightarrow 15x - 3y = 45$

M1

Setting $x = 0$ (eg)

M1

Any correct point, eg $(0, -15, -25)$,
 $(3, 0, -4)$, $(\frac{23}{7}, \frac{20}{7}, 0)$ $(4, 5, 3)$ etc

A1

3

(ii) $\mathbf{r} = \begin{bmatrix} 3 \\ 0 \\ -4 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 5 \\ 7 \end{bmatrix}$

ft their common point and d.v. from (a)
 Penalise missing r or $\mathbf{r}$ here

B1

1

[13]

Q7.

(a) $\begin{bmatrix} 7 \\ 4 \\ 6 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} = 14 + 4 + 18 = 36 \Rightarrow Q \text{ in } \Pi$

B1

1

(b) $\begin{bmatrix} -7 \\ 5 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} = -14 + 5 + 9 = 0$

Shown

B1

Explanation that l is perpendicular to l 's normal $\Rightarrow l$ is parallel to l

B1

1

Alternative:

$$\begin{bmatrix} 20-7\mu \\ 5\mu-8 \\ 3\mu+1 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} = 35 \quad (+0\mu)$$

(B1)

Explanation that, since this $\neq 36$ (and constant), l does not intersect l and must therefore be parallel to it

(B1)

2

(c) **Mark according to whichever scheme gives the greatest credit**

MARK SCHEME I

For $\mathbf{p} = \mathbf{q} + \lambda \mathbf{n} = \begin{bmatrix} 7+2\lambda \\ 4+\lambda \\ 6+3\lambda \end{bmatrix}$ for some λ

B1

For $PQ = |\lambda|\sqrt{14}$ or equivalent

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$$\overrightarrow{PR} = \mathbf{r} - \mathbf{p} = \begin{bmatrix} 20-7\mu \\ 5\mu-8 \\ 3\mu+1 \end{bmatrix} - \begin{bmatrix} 7+2\lambda \\ 4+\lambda \\ 6+3\lambda \end{bmatrix}$$

M1

$$= \begin{bmatrix} -7\mu-2\lambda+13 \\ 5\mu-\lambda-12 \\ 3\mu-3\lambda-5 \end{bmatrix}$$

A1

Setting $\overrightarrow{PR} \cdot \begin{bmatrix} -7 \\ 5 \\ 3 \end{bmatrix} = 0$

M1

Solving linear equation for λ

$$\Rightarrow 83\mu - 166 = 0 \text{ ultimately}$$

M1

$$\mu = 2 \text{ and/or } \overline{PR} = \begin{bmatrix} -(2\lambda+1) \\ -(\lambda+2) \\ 1-3\lambda \end{bmatrix}$$

A1

$$PR^2 = PQ^2$$

$$(1+2\lambda)^2 + (2+\lambda)^2 + (1-3\lambda)^2 = 14\lambda^2$$

M1

$$(\lambda = -3)$$

$$P = (1, 1, -3)$$

A1

9

MARK SCHEME II

$$\text{For } \mathbf{p} = \mathbf{q} + \lambda \mathbf{n} = \begin{bmatrix} 7+2\lambda \\ 4+\lambda \\ 6+3\lambda \end{bmatrix} \text{ for some } \lambda$$

B1

$$\text{For } PQ^2 = 14\lambda^2$$

B1

$$\overline{PR} = \mathbf{r} - \mathbf{p} = \begin{bmatrix} 20-7\mu \\ 5\mu-8 \\ 3\mu+1 \end{bmatrix} - \begin{bmatrix} 7+2\lambda \\ 4+\lambda \\ 6+3\lambda \end{bmatrix}$$

M1

$$= \begin{bmatrix} -7\mu-2\lambda+13 \\ 5\mu-\lambda-12 \\ 3\mu-3\lambda-5 \end{bmatrix}$$

A1

$$PR^2 = (13+7\mu+2\lambda)^2 + (12-5\mu+\lambda)^2 + (5-3\mu+3\lambda)^2$$

M1

$$\text{Setting } PR^2 = PQ^2$$

M1

$$\Rightarrow 83\mu^2 - 332\mu + 338 + 2\lambda = 0$$

Correct quadratic in μ

A1

Considering discriminant = 0 or $83(\mu - 2)^2 = -2(\lambda + 3)$

M1

$$\mu = 2, \lambda = -3 \text{ and } P(1, 1, -3)$$

A1

9

MARK SCHEME III

For $\mathbf{p} = \mathbf{q} + \lambda \mathbf{n} = \begin{bmatrix} 7+2\lambda \\ 4+\lambda \\ 6+3\lambda \end{bmatrix}$ for some λ

B1

For $PQ^2 = 14\lambda^2$

B1

For $\mathbf{r} = \begin{bmatrix} 20-7\mu \\ 5\mu-8 \\ 3\mu+1 \end{bmatrix}$

B1

For $QR^2 = (13 - 7\mu)^2 + (5\mu - 12)^2 + (3\mu - 5)^2$
 $= 83\mu^2 - 332\mu + 338 = 83(\mu - 2)^2 + 6$

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M1

For R closest to Q when $\mu = 2$,

M1

$$R = (6, 2, 7)$$

A1

Then $\overline{RP} = \begin{bmatrix} 1+2\lambda \\ 2+\lambda \\ 3\lambda-1 \end{bmatrix}$
ft

B1

Setting $PR^2 = PQ^2 \Rightarrow (1 + 2\lambda)^2 + (2 + \lambda)^2 + (3\lambda - 1)^2 = 14\lambda^2$

M1

$$(\lambda = -3) \quad P = (1, 1, -3)$$

A1

9

[12]

Q8.

(a) $\det \mathbf{A} = 5p - 1$

B1

$$\det \mathbf{B} = p^2 - 10p - 11$$

M1A0 if num error(s) made

M1A1

3

(b) Use of $\det(\mathbf{AB}) = \det \mathbf{A} \det \mathbf{B}$
PI

B1

Finding three values of p

Allow correct factors here

M1

$$p = \frac{1}{5}, 11, -1$$

ft numerical errors in (a)

A1F

3

[6]

Q9.

(a) **Either** $\begin{bmatrix} 5 \\ 3 \\ 4 \end{bmatrix} \times \begin{bmatrix} 1 \\ 6 \\ 2 \end{bmatrix} = -3 \begin{bmatrix} 6 \\ 2 \\ -9 \end{bmatrix}$

Correct vector product only here

M1A1

Explaining that d.v. of L is in dirn. of Π 's nml. $\Rightarrow L \perp \Pi$

B1

Or $\begin{bmatrix} 6 \\ 2 \\ -9 \end{bmatrix} \cdot \begin{bmatrix} 5 \\ 3 \\ 4 \end{bmatrix} = 0$ and $\begin{bmatrix} 6 \\ 2 \\ -9 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 6 \\ 2 \end{bmatrix} = 0$

(M1)(A1)

Explaining that d.v. of L is $\perp^r$ to 2 (non-//) vectors in $\Pi \Rightarrow L \perp^r \Pi$
Not just stating

(B1)

3

(b) E.g. $6 \times \textcircled{1} - \textcircled{2}$: $46 = 34p + 27q$
Eliminating r from any pair of eqns.

M1

$2 \times \textcircled{1} - \textcircled{3}$: $-57 = 21p + 6q$
 $\textcircled{2} - 3 \times \textcircled{3}$: $-217 = 29p - 9q$
Any 2 correct eqns (1 mark each)

A1A1

$2 \times \textcircled{4} + 9 \times \textcircled{5}$: $605 = -121p$
Solving a 2×2 system (any means) in order to get values for p, q, r

M1

$p = -5, q = 8, r = -1$

All 3 ✓ CAO

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A1

5

(c) $7 + 6t = -2 + 5\lambda + \mu$
(i) $8 + 2t = 0 + 3\lambda + 6\mu$
 $50 - 9t = -25 + 4\lambda + 2\mu$
Including re-arrangement attempt

M1

$9 = -6t + 5\lambda + \mu$
 $\rightarrow 8 = -2t + 3\lambda + 6\mu$
 $75 = 9t + 4\lambda + 2\mu$
i.e. the above system with
 $p = -t, q = \lambda$ and $r = \mu$

A1

2

(ii) Substg. $t = 5$ into L 's eqn.

Or $\lambda = 8$ and $\mu = -1$ into l 's eqn.

M1

$$P = (37, 18, 5)$$

CAO

A1

2

[12]

Q10.

(a) $x - 2y = -1 - \lambda$
 $-x + y = 3 - 3\lambda$

At least one correct from setting $z = \lambda$

B1

Solving for x and y in terms of λ

M1

$$x = 7\lambda - 5 \text{ and } y = 4\lambda - 2$$

CAO

A1

3

(b) Subst^g. x, y, z in terms of λ in $5x + ky + 17z = 1$

M1

$$35\lambda - 25 + k(4\lambda - 2) + 17\lambda - 1 = 0$$

$$\text{Factsn. attempt: } (4\lambda - 2)(k + 13) = 0$$

dM1

$$(2\lambda - 1)(k + 13) = 0$$

ANSWER GIVEN

A1

3

(c) (i) When $k = -13$, $5x - 13y + 17z$
 $= 35\lambda - 25 - 52\lambda + 26 + 17\lambda \equiv 1$

Subst^g. into 3rd eqn. and demonstrating consistency

B1

The three planes intersect in a line

B1

Solns. $x = 7\lambda - 5$, $y = 4\lambda - 2$, $z = \lambda$
ft

B1F

(ii) When $k \neq -13$, $\lambda = \frac{1}{2}$

B1

Soln. $(-1\frac{1}{2}, 0, \frac{1}{2})$
ft

B1F

Three planes meet at a point

B1

6

[12]

Q11.

(a) Use of $\cos \theta = \frac{\text{scalar product}}{\text{product of moduli}}$

Must be $\begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix}$ and $\begin{bmatrix} 5 \\ 1 \\ -1 \end{bmatrix}$

"sin θ =" scores M0 at this stage

M1

Numerator = 7 Denominator = $\sqrt{21}\sqrt{27}$

Allow denominator unsimplified

B1,B1

$\theta = 72.9^\circ$

CAO (but A0 if candidate proceeds to find its complement)

A1

4

(b) (i) $a + 4b = 7$ and $a - b = 12$
At least one correctly stated

B1

Solving simultaneously

M1

$$a = 11 \quad \text{and} \quad b = -1$$

CAO

A1

3

$$(ii) \quad \begin{bmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 5 & 1 & -1 \\ 2 & 1 & 4 \end{bmatrix} = \begin{bmatrix} 5 \\ -22 \\ 3 \end{bmatrix}$$

*For any valid, complete method for finding a suitable direction **vector**, eg finding a 2nd common point, eg $(2\frac{1}{2}, 0, \frac{1}{2})$ or $(1\frac{2}{3}, 3\frac{2}{3}, 0)$, and then **dv** = difference*

M1

CAO

A1

2

$$(iii) \quad \mathbf{r} = \begin{bmatrix} 0 \\ 11 \\ -1 \end{bmatrix} + \lambda \begin{bmatrix} 5 \\ -22 \\ 3 \end{bmatrix}$$

Must be a line equation and use their (b)(ii)

M1

or other equivalent line form

eg $(\mathbf{r} - \mathbf{a}) \times \mathbf{d} = \mathbf{0}$

ft their (b)(i) point, or any other correct point on the line

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A0 if no $\mathbf{r} =$ or $\mathbf{l} =$ etc

A1

2

[11]

Q12.

(a) Use of $\cos \theta = \frac{\text{scalar product}}{\text{product of moduli}}$

Must be $(3\mathbf{i} - 4\mathbf{j} + \mathbf{k})$ and $(2\mathbf{i} + 12\mathbf{j} - \mathbf{k})$

M1

Numerator = ± 43

B1

Denominator = $\sqrt{26} \cdot \sqrt{149}$



$$Dr. = 5.099... \times 3.742... = 0.6908...$$

B1

$$\theta = 46.3^\circ$$

A1

4

(b) $3x - 4y + z = 2$ and $2x + 12y - z = 38$

B1 B1

2

(c) $(3\mathbf{i} - 4\mathbf{j} + \mathbf{k}) \times (2\mathbf{i} + 12\mathbf{j} - \mathbf{k})$

M1

$$= -8\mathbf{i} + 5\mathbf{j} + 44\mathbf{k}$$

A1

p.v. of any point on line

M1

e.g. $(0, 5, 22), (8, 0, -22), (4, 2\frac{1}{2}, 0)$

A1

$$\frac{x-x_c}{-8} = \frac{y-y_c}{5} = \frac{z-z_c}{44}$$

ft

B1F

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5

Or Adding $\Rightarrow 5x + 8y = 40$ (e.g.)

Eliminating one variable

(M1)

$$\frac{x-8}{-8} = \frac{y}{5} = \lambda \quad \text{Or} \quad \frac{x}{-8} = \frac{y-5}{5} = \mu$$

Parametrisation attempted

(dM1)

(A1)

$$x = 8 - 8\lambda, \quad x = -8\mu$$

$$y = 5\lambda \quad y = 5 + 5\mu$$

Substⁿ. to find third variable

(M1)

$$\Rightarrow z = 44\lambda - 22 \quad \Rightarrow z = 44\lambda + 22$$

$$\frac{x-x_c}{-8} = \frac{y-y_c}{5} = \frac{z-z_c}{44}$$

(A1)

(5)

[11]

Q13.

- (a) $\lambda = -4$ gives $P(-29, 42, -19)$ on l

Correct value of λ

B1

1

- (b) $\cos^{-1} \frac{1}{9}$ or 83.6° (or 84°) or 1.46 rads.

*ft from 3rd d.c. or by any other method
(e.g. scalar product)*

N.B. Mark lost if 6.4° is then offered as the answer

B1 ✓

1

- (c) (i) $\mathbf{n} = 3\mathbf{i} - 4\mathbf{j} + 5\mathbf{k}$

B1

1

- (ii) Use of $\cos \theta = \frac{\text{scalar product}}{\text{product of moduli}}$
Must be direction vector of l and their $\mathbf{n}$

M1

$$\text{Nr.} = 45 \quad \text{Dr.} = \sqrt{50.9}$$

ft the "9" if necessary from (b)(i)

A1 A1

$$\theta = 45^\circ$$

CAO

A1

4

- (d) Substⁿ. $\begin{pmatrix} 3+8\lambda \\ 26-4\lambda \\ \lambda-15 \end{pmatrix}$ in $3x - 4y + 5z = 100$

$$3(3 + 8\lambda) - 4(26 - 4\lambda) + 5(\lambda - 15) = 100$$

M1

Solving a linear eqn. in λ

dM1

$$\lambda = 6$$

CAO

A1

$$\Rightarrow Q = (51, 2, -9)$$

ft their λ in l

B1 ✓

4

(e) $PQ = \sqrt{80^2 + 40^2 + 10^2} = 90$
ft

B1

Sh. Dist. = $90 \sin 45^\circ = 45\sqrt{2}$ or 63.6(4...)

M1

ft

A1 ✓

3

Or $p + m$ n subst^d. into $\Pi \Rightarrow m = 9$

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(M1)

$$\Rightarrow R = (-2, 6, 26)$$

$R =$ foot of perp^r. from P to Π

(A1)

$$PR = \sqrt{27^2 + 36^2 + 45^2} = 45\sqrt{2}$$

ft

(B1 ✓)

(3)

[14]

Q14.

(a) (i) $\mathbf{n} = (3\mathbf{j} + \mathbf{k}) \times (4\mathbf{i} - \mathbf{j}) = \mathbf{i} + 4\mathbf{j} - 12\mathbf{k}$

M1A1

2

Alternative:

$$\begin{bmatrix} 0 \\ 3 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} a \\ b \\ c \end{bmatrix} = 0 \quad \text{and/or} \quad \begin{bmatrix} 4 \\ -1 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} a \\ b \\ c \end{bmatrix} = 0$$

(M1)

$$\Rightarrow c = -3b, b = 4a \text{ i.e. } a(\mathbf{i} + 4\mathbf{j} - 12\mathbf{k})$$

(A1)

(2)

(ii) $d = (2\mathbf{i} + 5\mathbf{j} + \mathbf{k}) \cdot (\mathbf{i} + 4\mathbf{j} - 12\mathbf{k}) = 10$

Their n

M1A1 ✓

2

(b) Either substituting.

$$r = \begin{bmatrix} 2+3t \\ 2 \\ 5-t \end{bmatrix} \text{ into } \left(r = \begin{bmatrix} -1 \\ 2 \\ 6 \end{bmatrix} \right) \times \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix}$$

M1

And showing result is 0

A1

2

Or

Explaining p.v. and d.v. of line

Give the M1 for either

M1A1

2

(c) Substⁿ. $r = \begin{bmatrix} 2+3t \\ 2 \\ 5-t \end{bmatrix}$ into their plane eqn.

M1

Solving a linear eqn. in t

$$2 + 3t + 8 - 60 + 12t = 10$$

dM1

$$t = 4$$

A1 ✓

Pt. of intersection at $14\mathbf{i} + 2\mathbf{j} + \mathbf{k}$
CAO

B1

4

First alternative (c)

Uses (b)'s given form for L and the original form for Π

Setting
$$\begin{bmatrix} 2+3t \\ 2 \\ 5-t \end{bmatrix} = \begin{bmatrix} 2+4\mu \\ 5+3\lambda-\mu \\ 1+\lambda \end{bmatrix}$$

Must make an attempt to solve

(M1)

$$\mu = \frac{3}{4}t, \mu = 3+3\lambda, \lambda = 4-t$$

(A1)

Eliminating λ and μ (e.g.) $\Rightarrow t = 4$

(dM1)

Pt. of i/sctn. at $14\mathbf{i} + 2\mathbf{j} + \mathbf{k}$
CAO

(A1)

4

Second alternative (c)

Uses other given form for L

$$\left(\begin{bmatrix} 2+4\mu \\ 5+3\lambda-\mu \\ 1+\lambda \end{bmatrix} - \begin{bmatrix} -1 \\ 2 \\ 6 \end{bmatrix} \right) \times \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix} \text{ gives}$$

$$\begin{bmatrix} 3+4\mu \\ 3+3\lambda-\mu \\ \lambda-5 \end{bmatrix} \times \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} -3-3\lambda+\mu \\ 3\lambda+4\mu-12 \\ 3\mu-9\lambda-9 \end{bmatrix}$$

(M1 A1)

Equating to $\mathbf{0}$ and solving

$$\Rightarrow \lambda = 0, \mu = 3$$

(M1)

Pt. of intersection at $14\mathbf{i} + 2\mathbf{j} + \mathbf{k}$

CAO

(A1)

4

[10]

Q15.

(a) $\cos \theta = \frac{\text{scalar product}}{\text{product of moduli}}$
Must be the two normals

M1

Scalar product in numerator = 24

B1

Both moduli in denominator: $\sqrt{50}$ and $\sqrt{18}$

B1

$\cos \theta = \frac{4}{5}$ or 0.8

A1

4

(b) (i) $\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 4 & 5 & 3 \\ 4 & 1 & 1 \end{vmatrix} = 2\mathbf{i} + 8\mathbf{j} - 16\mathbf{k}$

M1

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Or via two scalar products, etc.

A1

2

(ii) $\mathbf{r} = \lambda \begin{bmatrix} 1 \\ 4 \\ -8 \end{bmatrix}$ or $\mathbf{r} \times \begin{bmatrix} 1 \\ 4 \\ -8 \end{bmatrix} = \mathbf{0}$
ft d.v. (any non-zero multiple)

B1ft

Through O

B1

2

*Must be a line equation.
 [Give 1/2 for correct cartesian equation.]*

[8]

Examiner reports

Q3.

Generally there was good response, with all students able to score marks in the earlier part of the question. The vector product evaluation through a determinant is well known; nevertheless, a handful of students made sign errors. Students also know how to apply the determinant to finding the area of a parallelogram. Many students did not appreciate the link between the parts of this question. Thus the determinant evaluated in part (a) was re-evaluated in part (b) albeit with the diagonals of the parallelogram instead. With regard to vector notation, students should realise that they must make key vectors clear, eg the

$$\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

zero vector written as $\mathbf{0}$ or $\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$. Many students did not find the coordinates of $\mathbf{M}$ by using the mid-point result from Core 1 but instead chose to find the intersection point of two lines. Part (c) proved most challenging. Those who used the scalar product or Cartesian form of the plane achieved the greatest success in finding the point of intersection of the line and plane.

Q4.

This topic of equations of lines and planes is not well understood by students. Surprisingly, part (a) was not always answered correctly with arguments concerning the scalar product often wrongly used. Part (b) was most often successful when two points on the line were used or if the scalar product of the normal to the plane and direction vector of the line was set equal to zero and solved to find p . Part (c)(i) proved very discriminating with students using both scalar and vector products in tandem but not using the correct trigonometric ratio and confusing sine and cosine.

Q5.

This was another high-scoring question for many candidates, especially since the three parts were essentially completely independent of each other's results. Part (a) was straightforward and routine, though rather a lot of candidates lost marks by failing to give the acute angle required, subtracting the correct one from 90° , mixing up sines and cosines in the working, missing the minus sign in front of the "4" in the second of the two given normal vectors, or making numerical slips in adding up the three squares in each of the moduli.

Q6.

Responses to this question were usually much more confident and, apart from the usual batch of numerical and / or sign slips, decently correct. Finding points and directions proved to be much more what students were expecting.

Q7.

As already mentioned in the introduction, this question proved to be very demanding and frequently candidates offered no attempt. Even though the three marks available in parts (a) and (b) were relatively straightforward, hardly any students managed to acquire all of them.

This was because in part (b) almost all students undertook the expected method, gaining a zero scalar product, but then failed to consider that some explanation would be required as to why a successful perpendicularity test established parallelism.

Students arguably could be criticised for spurning some relatively straightforward marks earlier on in this paper, but not in the last part of this final question: it was always intended to discriminate and to stretch the most able. Most students did not know where to begin and so made no attempt. Most of the rest presented almost unintelligible amounts of any vector method they could think of at the problem, almost invariably without any explanation or structure to their working. As a result, there were three or four marks which we were prepared to give for useful introductory generalised points, vectors or distances, but hardly anyone could manage even these. It did not help when many, of the few who did try to do something here, thought that equal distances PQ and PR meant equal vectors $\overrightarrow{PQ}$ and $\overrightarrow{PR}$.

Q8.

Most candidates scored highly on this question. In part (b), almost all candidates used the product of the two determinants found in part (a), but a substantial number showed a lack of maturity by expanding the product into a cubic expression and then attempting to factorise it, not always correctly.

Q9.

There were several easy marks to be had in this question, and the rest was helpfully sign-posted. Part (a) was routine ... or it would have been had not, again, some sort of explanation been required for full marks. Approaches were evenly split between using the scalar-product and the vector-product. In the former case, a lot of candidates seemed to think they only needed to show that the line's direction vector was perpendicular to **one** vector parallel to the plane. In the second case, many candidates were content merely to find the vector product of the two vectors "in" (parallel to) the plane; others noted that it was a multiple of the d.v. of L without further comment. For the 3rd mark on this part of the question, we wanted an explanation as to *why* their scalar- or vector-product work established the given outcome – it is not enough for candidates merely to repeat the result that we gave them. *Given* results always require more of candidates by way of 'dotting the i's and crossing the t's'.

The more discerning teachers and students of MFP4 will understand why part (b) was set up as it was, since this is the only way to force candidates not just to resort to calculator-output generated answers. In part (c), remarkably few got past equating the equations of L and II ; the majority of candidates had no real idea what to do and so there were lots of answers seen in which the point of intersection was taken to be $(-5, 8, -1)$.

Q10.

This question was definitely the most demanding on the paper; partly because this topic of solving systems of equations has generally proved to be one of the least palatable topics to candidates over the years, but mostly due to the poor examination technique employed on this occasion. Many candidates ignored the directly-stated demand to set $z = \lambda$ in part (a).

It was unfortunate that few candidates seemed to notice that the given result in part (b) should be used in part (c). This resulted in much unnecessary work for many candidates and often a failure to answer the question asked. The demand was for candidates to "determine" any solutions, and then to describe what they were in relation to the planes. Many candidates ignored the first of these requests altogether, and then went on to describe the arrangement of the planes, which was not what the question said. A large majority of candidates lost four or five marks, in part (c)(i) by failing to re-state their answer to part (a) and then calling it a "line" (of intersection) and in part (c)(ii) by failing to

notice the consequence of part (b)'s answer that $z = \lambda = \frac{1}{2}$, which then gives a point (of intersection) whose coordinates can be written straight down (again, using part (a)'s answer).

Q11.

This proved to be another accessible question, with the question's structuring helping candidates. Only a few opted to start again for part (b)(iii), for instance.

Q12.

This was the longest question on the paper, and contained a mixture of ideas from across the unit's topics. Most candidates coped very favourably with this variety and, minor arithmetical slips apart, marks were high. The biggest obstacle to success came in part (b)(i), where candidates needed to find the coordinates of any one point on the line of intersection. Setting x , y or z equal to zero and finding a value for the other two variables from the resulting simultaneous equations was the most common approach, and this worked well for most candidates.

Q13.

Parts (a) to (d) of this question were usually very well done, and candidates were able to score a lot of the marks. The big surprise came in part (e) when so few arrived at the correct final answer. In almost all cases, a simple diagram would have helped enormously, and they would then have seen that a simple bit of basic trigonometry would have done the trick. In most cases, candidates seemed to be working with randomly-selected vectors from the first few parts of the question and trying to insert them into some (often half-) remembered vector formula.

Q14.

This was one of the questions which candidates with a weak grasp of the underlying ideas behind topics found difficult. Part (a)(i) can be done either by a single vector product or by two scalar products. A large number of candidates then used one of the plane's two direction vectors, rather than the position vector of the point $(2, 5, 1)$ which lies in the plane Π , to find the scalar constant d required. This is one of several fundamental errors made by candidates.

Part (b) was a straightforward test of the alternate form of a line equation, yet it was frequently answered in unconvincing ways. The instruction to "verify" – that is, to substitute the given expression for $\mathbf{r}$ into the given equation for L – was clearly not understood by many. The perfectly allowable alternative approach is to explain the role of the line's direction vector and show that the point $(2, 2, 5)$ is a point on the line. Many candidates managed one or the other but there was often a confusing amount of vector algebra and arithmetic presented with no apparent purpose and absolutely no explanation. Such vague approaches usually gained one of the two marks, but not both.

Part (c) was generally done quite well, as there are at least three approaches, at least one of which does not require successful progress in any previous part of the question.

Q15.

This starter question was handled very capably on the whole. In part (a), many candidates felt compelled to work out the angle itself, even though the question explicitly asked for just the cosine of the angle. Large numbers of candidates went on to find some other

angle as well. Uncancelled rational answers, such as $24/30$, were acceptable here, but answers with uncancelled surds appearing were not awarded full marks. In part (b) (ii), very many candidates failed to see that the origin was a common point on the two given planes, and so spent some considerable time trying to find some other common point. A frequent error in this part was to give the equation of a plane rather than of a line.



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